

Qubit-Efficient QUBO for Interbank Payment Settlement and Counterparty Netting

1. Problem Statement:

In RTGS systems, payment gridlock occurs when banks lack individual liquidity to settle obligations resolvable through multilateral netting. The formulation captures counterparty netting naturally: incoming settled payments increase a bank's effective liquidity, enabling settlements that would otherwise be blocked. We model this as binary optimisation over a directed payment network and encode it as a QUBO.

2. QUBO Encoding:

We index payment arcs by k to handle multigraphs. We use the standard squared-penalty method (Glover et al., 2022). For each node, binary-encoded slack converts the inequality to equality.

The QUBO is $Q(x, s) = \sum_k w_k x_k - \sum_u \lambda_u h_u^2$.

Penalty weights $\lambda_u = \gamma \cdot \Sigma w_k$, validated by a γ -sweep. A Two-Phase SA searches only logical variables, computing optimal slack analytically.

3. Data:

Three synthetic interbank networks of increasing size (5, 6, 7 banks) with varied liquidity conditions (30%, 35%, 20%), plus a 6-bank gridlock scenario for IQPMS testing.

Datasets provided as CSV.

	Exp 1	Exp 2	Exp 3
Banks	5	6	7
Payments	8	12	14
Total obligations	40	50	93
System liquidity	14	20	20

4. Results:

All Q-matrices verified by exhaustive enumeration. SA finds the exact optimum in all three experiments.

Metric	Exp 1	Exp 2	Exp 3
True optimum	32	50	66
SA on QUBO	32 (100%)	50 (100%)	66 (100%)
Greedy sequential	2 (6%)	22 (44%)	7 (11%)
Greedy with netting	2 (6%)	50 (100%)	7 (11%)
Feasibility rate	9.4%	20.9%	0.2%
Standard qubits	28	34	47
IQPMS qubits	10	17+	19+
QUBO verified	PASS	PASS	PASS

Percentages are of the feasible optimum, not total obligations. At 20% liquidity (Exp 3), 27 of 93 obligations cannot settle; a genuine constraint, not an SA failure. IQPMS qubit counts use Table II of De Santis et al. for IN/OUT master slack and our strict IQP test for satellite slack.

5. Iteration: IQP and IQPMS Investigation:

65–71% of qubits are slack. At real RTGS scale (~50 banks, payments in millions), slack rises to ~93%. We investigated the IQP method (De Santis et al., 2026) for slack-free encoding. Strict IQP ($P=0$ on all satisfying, $P \leq -1$ on violating) is infeasible at nodes where the satisfying assignments span the full quadratic polynomial space (rank = n_{params}), which occurs at most nodes with $M \geq 3$. The paper's MS decomposition resolves this by introducing IN/OUT as master. The satellite (liquidity) then operates on a reduced satisfying set and achieves exact separation with 0 slack for nodes with $M \leq 5$. The master needs 0–2 slack per node (Table II). In all experiments, the general-settlement optimum already satisfies IN/OUT, confirming the cycle restriction imposes no cost under gridlock.

6. IQPMS Q-Matrix Implementation:

We implemented the full IQPMS Q-matrix for a 6-bank gridlock scenario (9 arcs, 25% liquidity). The IQP conditions (Eqs. 8–10) are solved numerically for all node topologies using LP-based coefficient minimisation. The IN/OUT master uses the paper's closed-form for $M \leq 3$ and numerical IQP for $M \geq 4$. Result: 13 total variables (9 logical + 4 slack)

vs 34 standard (9 + 25 slack) — an 84% slack reduction. Exhaustive verification confirms QUBO optimum = true optimum = 54. SA on the IQPMS formulation finds the true optimum; SA on the standard encoding reaches only 50. The smaller search space improves both classical and (expected) quantum solver performance. Full implementation in supplementary script `iqpms_qmatrix.py`.

7. Limitations:

Scale: largest instance is 14 payments (47 qubits). Static: single-point optimisation. IQPMS requires the IN/OUT constraint (gridlock-cycle assumption).

8. References:

1. De Santis, D., Tirone, S., Marmi, S., & Giovannetti, V. (2026). Optimized QUBO formulation methods for quantum computing. *Quantum Science and Technology*, 11, 015056. Also available as arXiv:2406.07681v2.
2. Glover, F., Kochenberger, G., Hennig, R., & Du, Y. (2022). Quantum bridge analytics I: a tutorial on formulating and using QUBO models. *Annals of Operations Research*, 314(1), 141–183.
3. Lucas, A. (2014). Ising formulations of many NP problems. *Frontiers in Physics*, 2, 5.
4. Bech, M. L. & Soramäki, K. (2001). Gridlock resolution in interbank payment systems. *Bank of Finland Discussion Papers*.